# Sharp server heuristic: influence and finite-breakdown characterization I

The server-side `robust_aggregate` rule is the sharp heuristic axis of the three-axes design (sec. 7.2.1). It has BH-rejected positive contrasts in the configured accuracy verdict in sec. 7.5.1 but has declared reversals elsewhere. Unlike the objective-backed `variational_aggregate`, no closed-form free-energy derivation has been established for it in this repository. A separate scoped proposition in the aggregation-objective supplement rules out the declared continuously differentiable, separable forward-KL objective class for the implementation's raw log-pool block; it does not rule out every broader coupled or fixed-point-only construction. The rule therefore remains a heuristic whose positive formal property is bit-identical recovery of the log-linear pool at `robustness = 0` (eq. 7). This section does not promote the scoped negative result into an objective certificate; it *measures* the heuristic empirically, and the measurement makes its honesty boundary concrete.

We measure two things (fig. 1). First, a **numerical influence function**: we drag one agent's belief a growing fraction toward a confident-wrong contamination point and read its converged pooling weight. At `robustness = 0` the weight is a flat $1/n$ at every perturbation — the naive pool never down-weights anyone — which anchors the instrument to the proven recovery corner. At positive robustness the dragged agent's influence falls (not strictly monotonically — a tiny drag can briefly *raise* it before the divergence penalty dominates, an honest non-monotonicity we report rather than smooth away).

Second, and more consequentially, a **breakdown witness**. We add colluding confident-wrong adversaries — all broadcasting the same false state — to a fixed colony of 5 honest sentinels until each aggregator's consensus argmax is *captured* (flips to the adversaries' target). The sharp heuristic is captured by 2 colluders; the conservative objective-backed variational rule withstands more, capitulating only at 4. Both counts are **finite** (Yes): a colluding majority overwhelms either rule. That finite breakdown point is the honest headline: neither rule has an unconditional truth-recovery claim under coordinated collusion. The absence of an objective theorem for `robust_aggregate` is a separate derivational boundary, and the finite capture measurement neither establishes estimator-level B-robustness nor refutes the variational rule's stated raw effective-weight result.

The report also runs a declared diagnostic grid over state dimension, honest-agent count, robustness, four simple attack mechanisms, and balanced versus adversary-downweighted base weights. This is a coverage instrument for finding counterexamples, not a random sample of worlds and not a theorem search over all simplexes. A finite capture row is evidence against a universal guarantee; an uncaptured row is only “not found within this search budget.”

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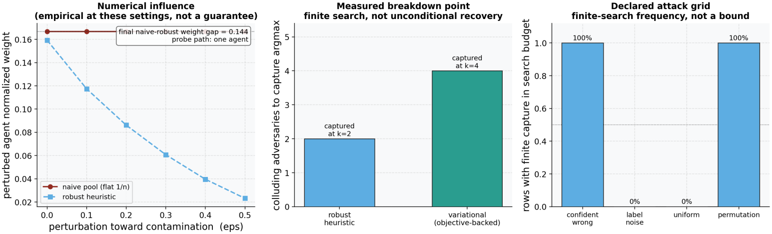


Figure 1: Source relation: original project diagnostic of the server-side heuristic; estimand: numerical influence, finite-search breakdown count, and declared-grid capture fraction; uncertainty: deterministic seeded colonies, so no resampling interval is shown. Empirical characterization of the `robust_aggregate` heuristic (two panels plus an optional attack-grid diagnostic). Left panel (numerical influence): the x-axis is the perturbation fraction by which one agent's belief is dragged toward a confident-wrong contamination point; the y-axis is that agent's converged normalized pooling weight, plotted for the naive pool (flat at $1/n$, dotted reference) and the robust heuristic (down-weighting). The inset reports the final naive-minus-robust weight gap at the end of the probed path. Labeled "empirical, at these settings — not a guarantee." Right panel (measured breakdown point): the x-axis is the aggregator (robust heuristic vs objective-backed variational); the y-axis is the number of colluding confident-wrong adversaries that captures that aggregator's consensus argmax — the robust heuristic at $k = 2$ and the variational rule at $k = 4$. Both bars are finite, so neither rule has an unconditional truth-recovery guarantee against coordinated collusion; this does not negate the variational rule's per-agent effective-weight theorem. Deterministic seeded colonies (no resampling), so no error band is applicable. The optional third panel reports the fraction of declared grid rows with finite capture within the configured adversary budget; it is not a probability or a global breakdown bound.